ARTIFICIAL NEURAL NETWORKS

HOW DO THEY WORK?

HOW NEURAL NETWORKS WORK

THOUGHT EXPERIMENT

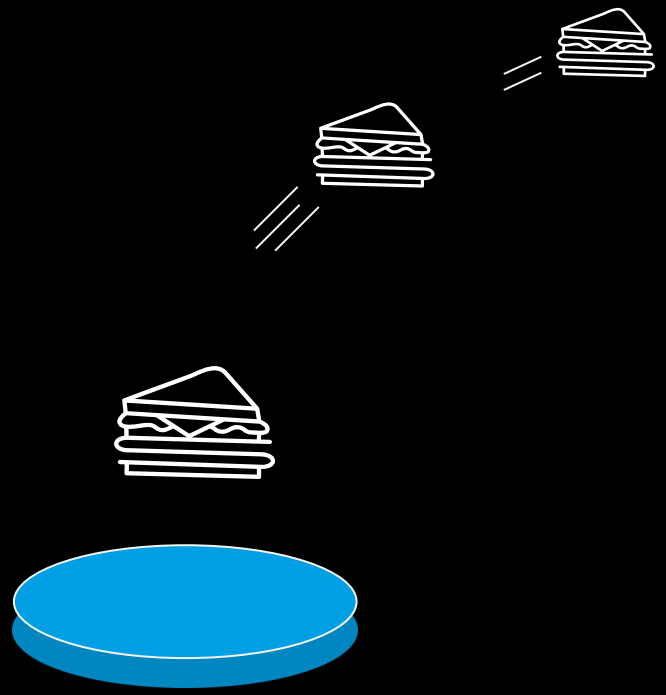
CREDITS: *I stole this example from the wonderful book "You Look Like a Thing and I Love You: How Artificial Intelligence Works and Why It's Making the World a Weirder Place" by Janelle Shane.*

A magic sandwich hole spits out random sandwiches with a finite set of possible ingredients:

- Cheese
- Eggshells
- Mud
- Chicken
- Peanut Butter
- Marshmallow

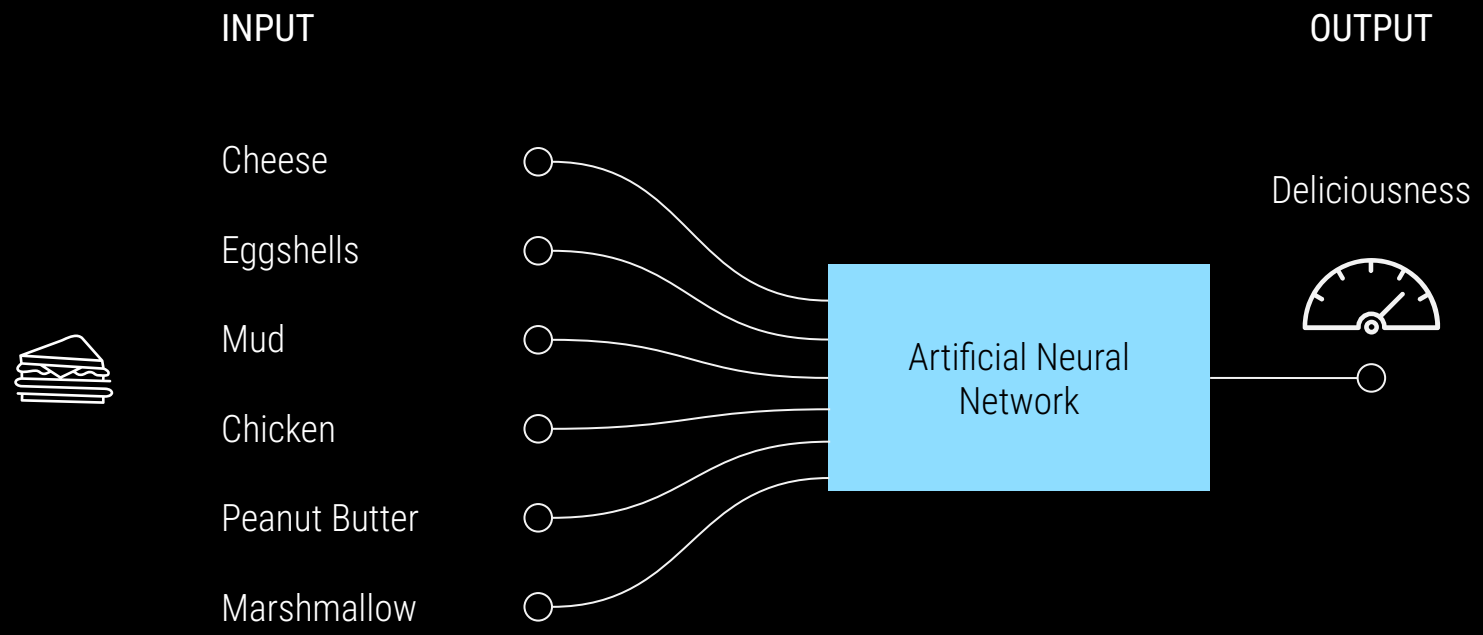
Only 1 in 1000 is actually good - but really, really good.

We want to train a neural network to identify the delicious ones!



HOW NEURAL NETWORKS WORK

PROBLEM STATEMENT



HOW NEURAL NETWORKS WORK

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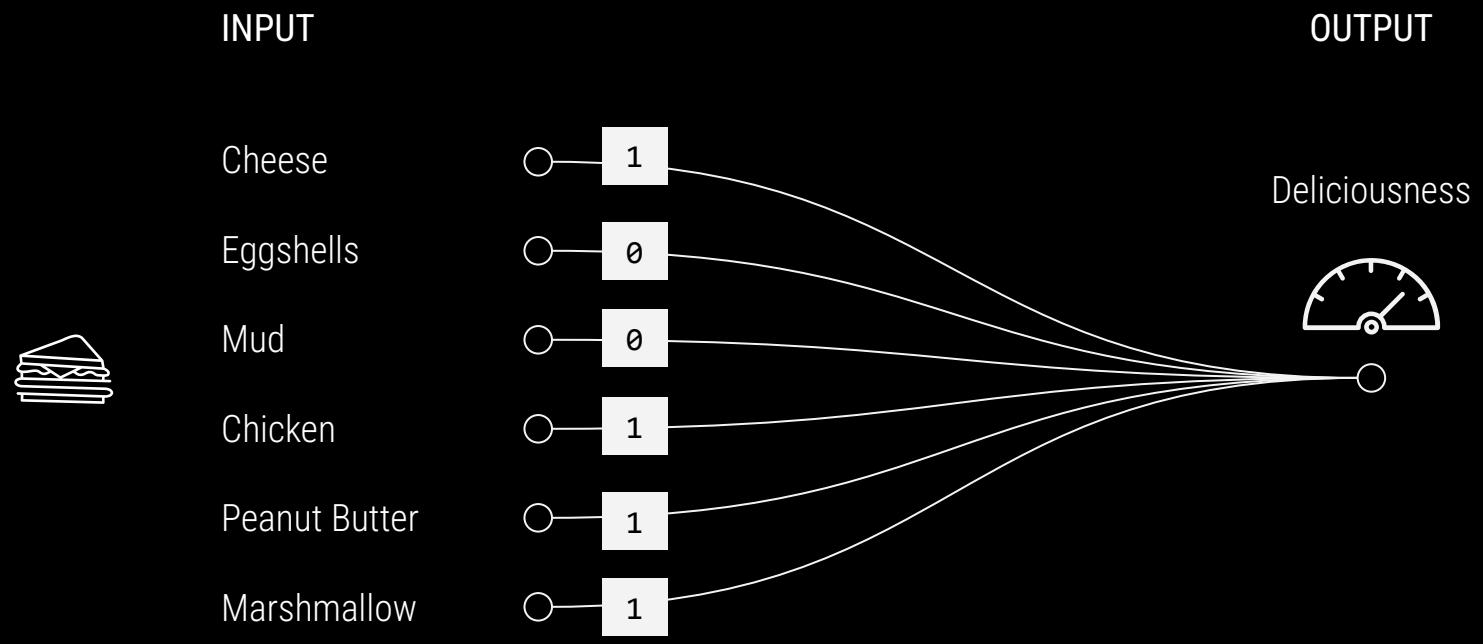


HOW NEURAL NETWORKS WORK

EGGSHELLS AND MUD?

INPUT

Cheese 1

Eggshells 0

Mud 0

Chicken 1

Peanut Butter 1

Marshmallow 1

OUTPUT

Deliciousness

= 0

Great :-)

HOW NEURAL NETWORKS WORK

PEANUT BUTTER AND MARSHMALLOW?

INPUT

Cheese 1

Eggshells 0

Mud 0

Chicken 1

Peanut Butter 1

Marshmallow 1

OUTPUT

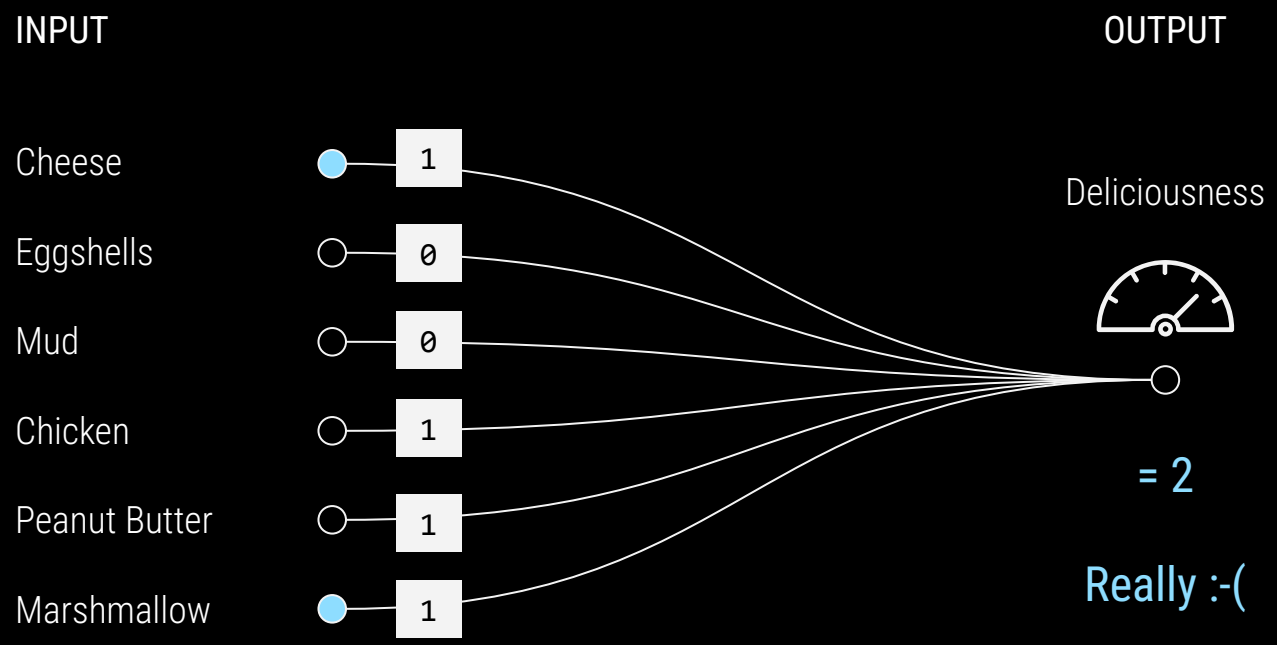
Deliciousness

= 2

Great :-)

HOW NEURAL NETWORKS WORK

CHEESE AND MARSHMALLOW?



HOW NEURAL NETWORKS WORK

ALL AT ONCE

INPUT

Cheese 1

Eggshells 0

Mud 0

Chicken 1

Peanut Butter 1

Marshmallow 1

OUTPUT

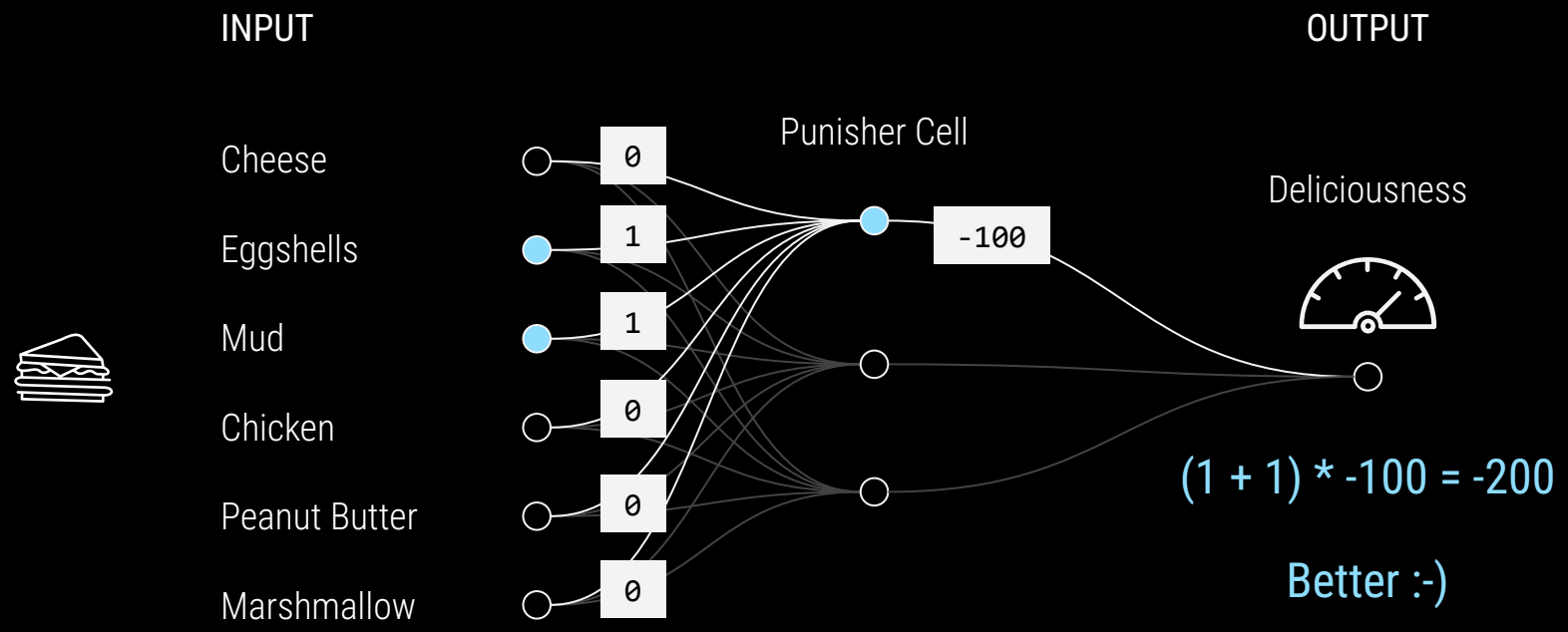
Deliciousness

= 4

Really ???

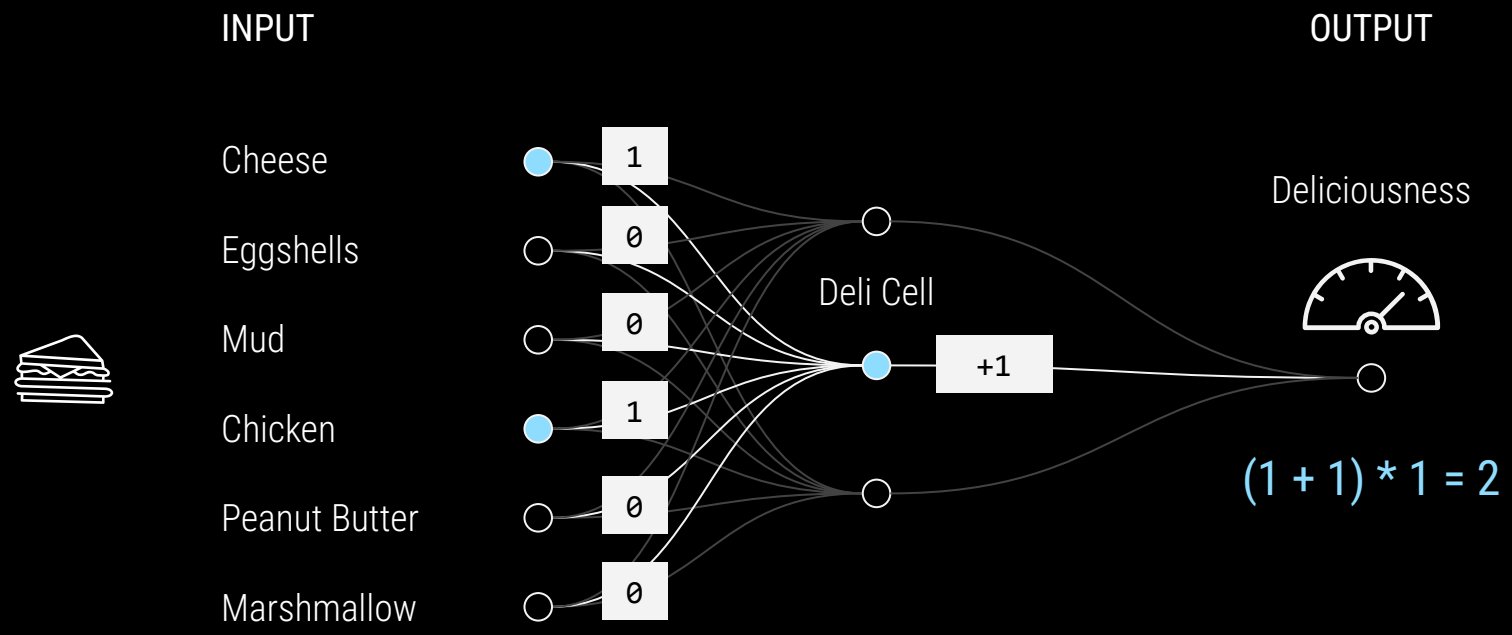
HOW NEURAL NETWORKS WORK

THE PUNISHER CELL



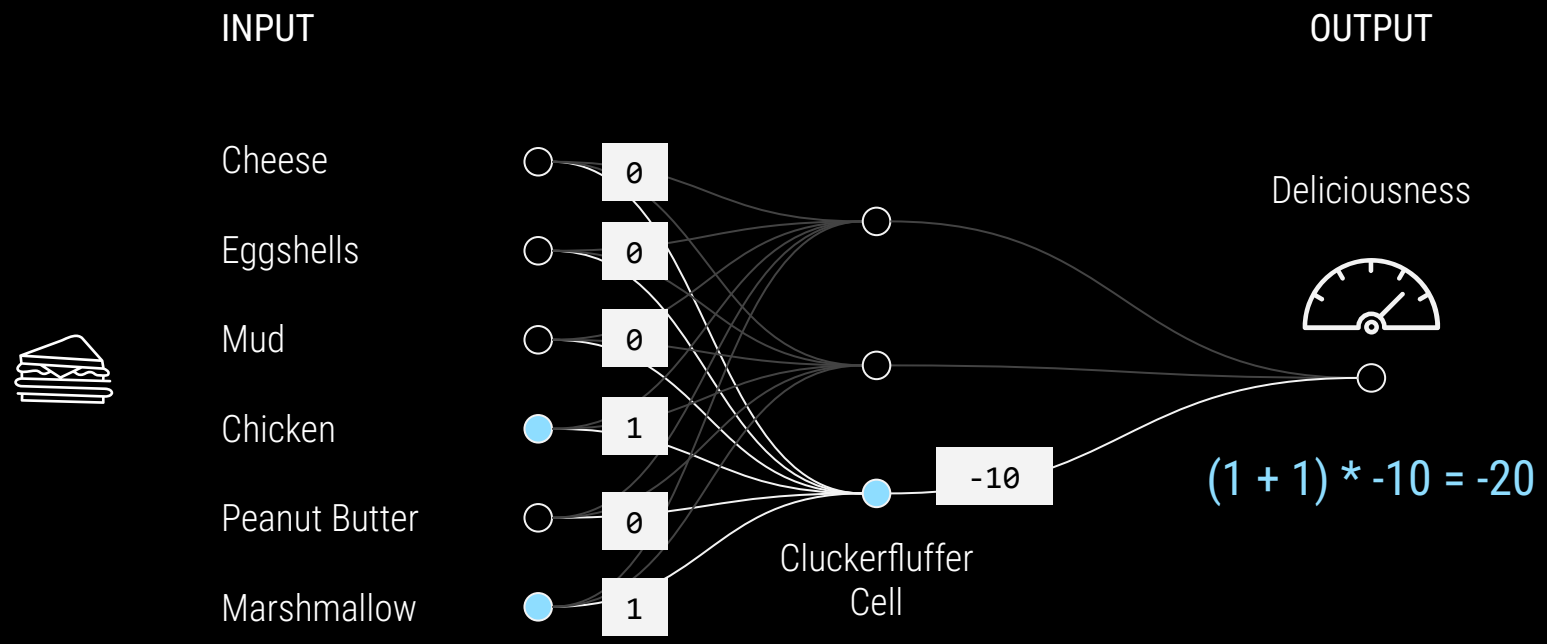
HOW NEURAL NETWORKS WORK

THE DELI SANDWICH CELL



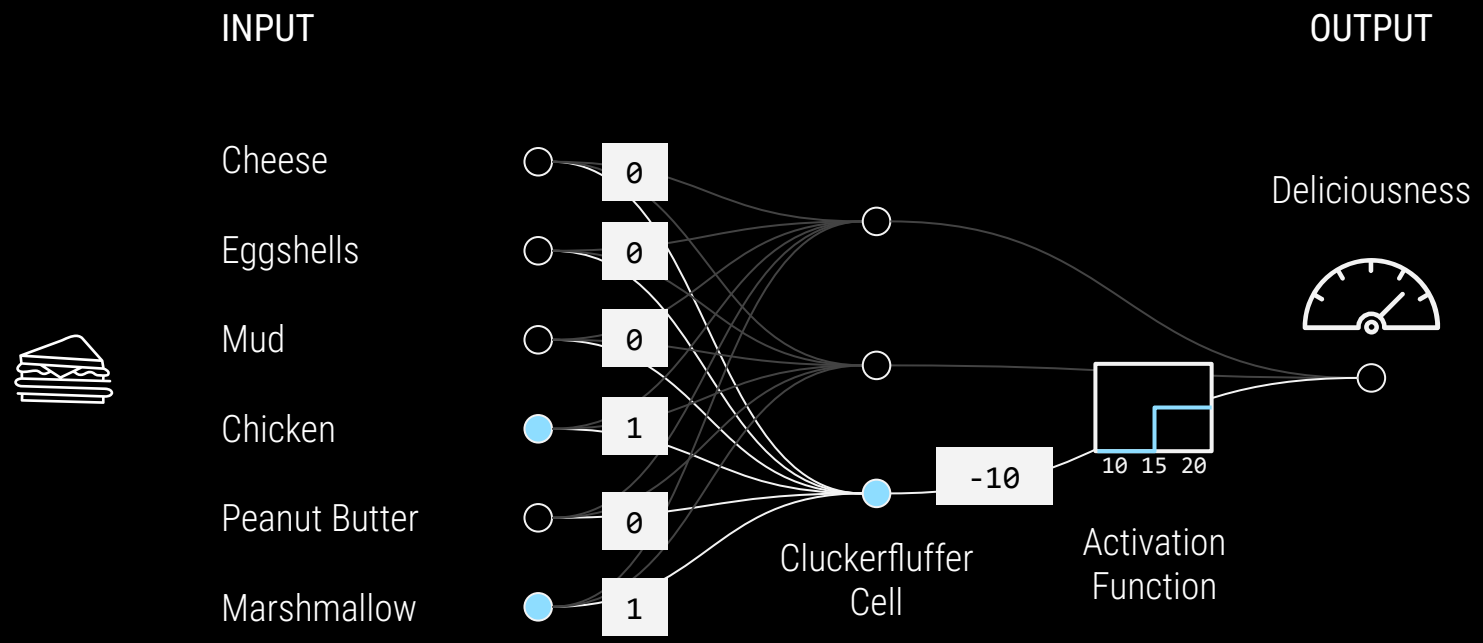
HOW NEURAL NETWORKS WORK

PUNISH CERTAIN COMBOS



HOW NEURAL NETWORKS WORK

THE ACTIVATION FUNCTION



HOW NEURAL NETWORKS WORK

TRAINING THE NEURAL NETWORK

Cheese	Eggshells	Mud	Chicken	Peanut Butter	Marshmallow	Delicious
1	0	0	1	0	0	9.5
0	1	1	0	0	0	0.0
1	0	1	0	1	0	0.5
1	0	0	0	0	0	6
0	0	0	0	1	1	8.9

HOW NEURAL NETWORKS WORK

TRAINING THE NEURAL NETWORK

This is the first record from the training data

The weights are randomly initialized

Cheese

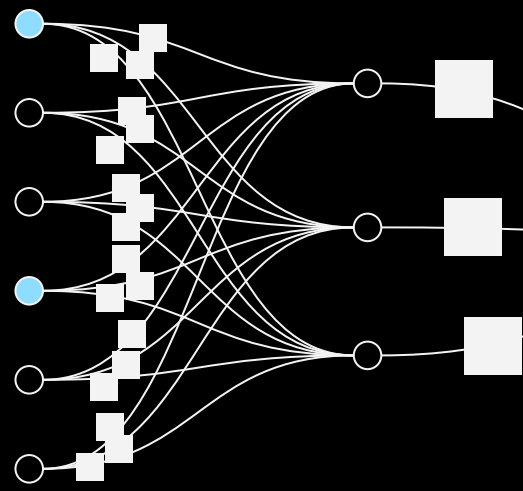
Eggshells

Mud

Chicken

Peanut Butter

Marshmallow



OUTPUT

Deliciousness

= -0.2 (for example)

This is likely to be nonsense

HOW NEURAL NETWORKS WORK

TRAINING THE NEURAL NETWORK

The weights are adjusted gradually, so the network is more likely to predict the correct score next time

Cheese

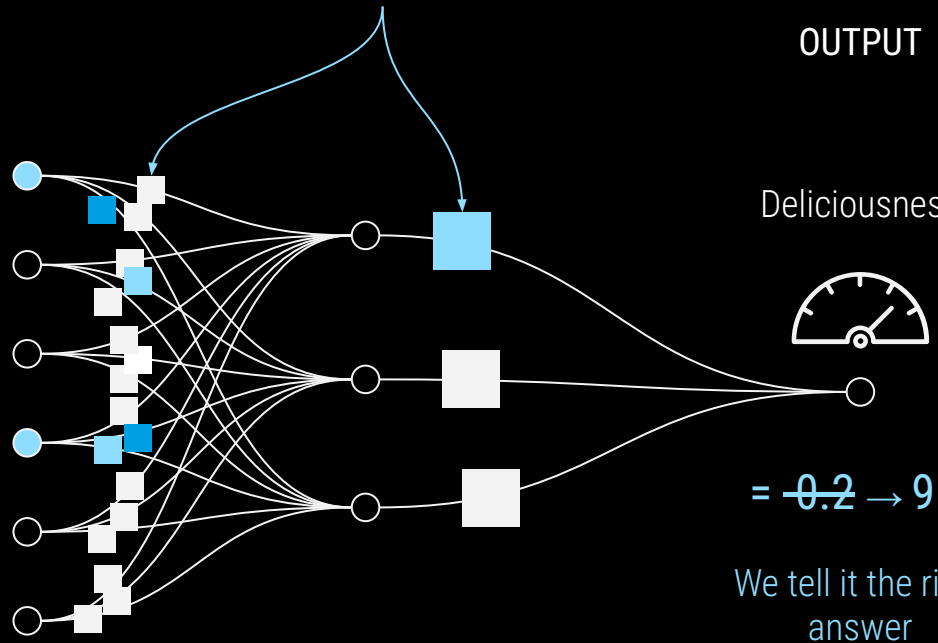
Eggshells

Mud

Chicken

Peanut Butter

Marshmallow



OUTPUT

Deliciousness

= ~~0.2~~ → 9.5

We tell it the right answer

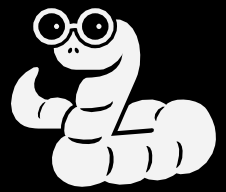
BRAIN COMPARISON

~~Researchers estimate ANNs could have as many neurons as a human brain by 2050 (~80 billion)~~

UPDATE: The GPT-3 neural network, released in May 2020, has **~170 billion** neurons.

UPDATE: GPT-4, released in March 2023, has ~1.2 trillion neurons.

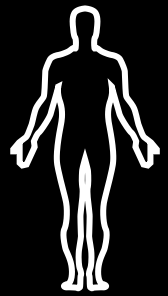
UPDATE: GPT-5, released in August 2025, is estimated to have 2-5 trillion neurons.

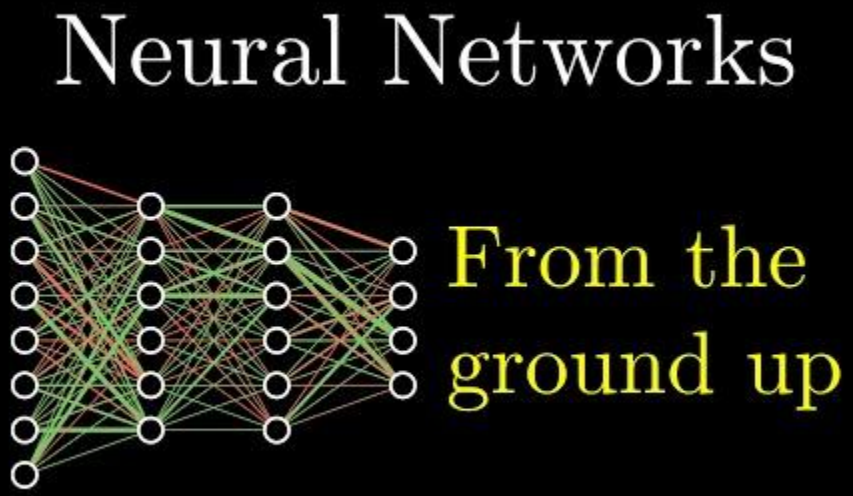


Modest sized ANNs have as many neurons as a worm's brain (~300)

Large ANNs come close to the number of neurons a bee's brain has (~170.000)

The largest ANNs have about the size of a frog's brain (16 million neurons)



A human brain has about 86 billion neurons

RECOMMENDED VIDEO